

First, we note four criteria that apply to each goal. At a high level, a good causal explanation explains the most general effects from the fewest non-trivial causes.⁹ Thus, the ideal causal mediator is one that achieves a Pareto optimum between maximal **sparsity** (using as little of the computation graph as possible), **generality** (explaining the widest breadth of data), **selectivity** (explaining only the phenomenon of interest and as little else as possible), and **faithfulness** (retaining fidelity to the original model).

Explaining model behavior. Often, our goal is to produce qualitative and human-understandable insights about how models perform a certain behavior or task. What makes a good explanation? In mechanistic interpretability, this can be defined as one that maximizes the above metrics of sparsity, generality, selectivity, and faithfulness, as well as **human understandability**. Faithfulness is already a common metric in mechanistic explanations (Hanna, Pezzelle, and Belinkov 2024; Marks et al. 2025; Wang et al. 2023); measuring it is straightforward, as it is a generalization of causal mediation analysis to evaluation.¹⁰ Sparsity is often called “minimality” (Mueller et al. 2025; Wang et al. 2023); this metric is not as common as faithfulness, but is becoming increasingly common in circuit analyses. It is typically either defined as the inclusion of as few redundant dependencies as possible (Wang et al. 2023), or as the inclusion of as few components as is necessary to achieve high faithfulness (Mueller et al. 2025). Generality is a rarer metric, as it requires evaluating on out-of-distribution examples that may need to be carefully curated. Recent work has argued that generality is a crucial metric, and has begun to define ways to measure it (Huang et al. 2025; Li et al. 2025). Selectivity is not often explicitly discussed nor measured, as it is assumed that maximizing faithfulness and sparsity should implicitly optimize selectivity; more work is needed to verify this assumption. The most difficult of these criteria to quantify is human understandability; well-trained sparse autoencoder features are typically easier to interpret than neurons, but it is not yet clear whether this metric can be measured in a reliable manner.

Verifying a mechanistic hypothesis. Sometimes, we already have a well-defined guess as to how a model accomplishes a task (a *mechanistic hypothesis*), and we would like to verify to what extent our hypothesis is accurate. Here, the criteria are similar as when explaining model behaviors: we would like to locate the smallest set of components or lowest-rank subspace that aligns best with the hypothesized explanation on the broadest possible data distribution. The primary difference is that the human-understandability of the mediator is of lesser importance for this goal, as understandability is a function of the human’s mechanistic hypothesis and its accuracy, rather than the method used to align the model with the hypothesized causal variables. Criteria for success thus include sparsity, generality, and selectivity, as well as a modified form of faithfulness that we term **counterfactual faithfulness**. Counterfactual faithfulness measures whether the model’s behavior changes in the expected manner when we perform counterfactual interventions to a specific part of the hypothesized mechanism; this is typically quantified as the interchange intervention accuracy (IIA; Geiger et al. 2021).

⁹ We add the “non-trivial” qualifier because one could trivially maximize the scope of an explanation by taking the entire computation graph $\mathcal{C}$ as a single mediator.

¹⁰ We refer readers to Hanna, Pezzelle, and Belinkov (2024) for further detail.

Localization and editing. We often simply want to know where in a model some ability is implemented without necessarily understanding the components that implement it. This enables applications like model editing, steering, and parameter-efficient model adaptation (e.g., with LoRAs applied to specific layers). Here, we want to maximize sparsity, generality, and selectivity, but we do not assign as great an importance to human understandability, and do not necessarily require a hypothesis for this to work well. If using localized components for an application like steering or model editing, then one should maximize primarily for **downstream task performance**, with sparsity and generality ideally being integrated into the downstream evaluation metrics. Note that high-quality evaluation is essential: one can maximize the efficacy of model editing via fine-tuning all model parameters, but this may produce unintended side effects if sparsity is ignored, and may not generalize well out-of-distribution if generality is ignored. Put simply, if there exist multiple mediator types that achieve similar test performance, then the best solution is likely that which then also attains the greatest sparsity and generality. Note that faithfulness is also important, but is not explicitly considered in model editing because it is trivially satisfied if the model’s behavior changes after the editing operation.

3. Related Work

Causal interpretability surveys do not always focus on model internals, and mechanistic interpretability surveys do not necessarily require causal grounding. We give a brief overview of both types of survey here, contrasting them with ours. We also discuss recent tooling efforts that have accompanied the growing interest in mechanistic interpretability.

Mechanistic/model-internal interpretability surveys. Some surveys catalogue studies that aim to understand the latent representations of neural networks (Belinkov and Glass 2019; Belinkov 2022; Sajjad, Durrani, and Dalvi 2022); these have often called for more causal validations of correlational observations. More recent surveys tend to focus increasingly on giving practical overviews of how to use common methods for intervening on model internals (Ferrando et al. 2024; Rai et al. 2024). Others provide perspectives for understanding the trajectory of the mechanistic interpretability field (Räuker et al. 2023), and/or cataloguing the impacts of the field (Bereska and Gavves 2024). Notably, each of these surveys taxonomizes the field based on methodological differences; for example, a common contrast might be circuit analysis (discovering causal graphs of task-specific causal dependencies from model components, as in Wang et al. 2023; Conmy et al. 2023) vs. causal variable alignment methods (aligning model representations to human-provided concepts, as in Geiger et al. 2021; Wu et al. 2023), even if they both operate over the same kinds of components.

There is a gap here: many of these methods implicitly deploy the same units of analysis, and thus benefit/suffer from the same fundamental pros/cons as a result. For example, it is not clear whether one would be better off discovering circuits over neurons, or attention heads, or abstractions over groupings of these. The same issue applies to any model-internals method. Thus, in this survey, we instead foreground the units of causal analysis that a study employs, as well as the way in which the study searches over those units, as primary factors in categorizing the study. We also ground the field in the language of causality, which grounds the goals of mechanistic interpretability: to discover causal subgraphs explaining how inputs are transformed into outputs.

Causal interpretability surveys. Moraffah et al. (2020) is a causal interpretability survey that categorizes various streams of causal interpretability research according to the methods they employ, though the studies they summarize are not necessarily based in the ideas of causal mediation analysis. The units of analysis were also not foundational to their organization, nor directly compared to each other. Other interpretability surveys (Subhash et al. 2022; Gilpin et al. 2018; Singh et al. 2024a) focus on methods for explaining the decisions of neural networks without causally grounding the explanation methods or focusing on model internals. Many causality-focused surveys are domain-specific, including areas such as cybersecurity (Rawal et al. 2024) and healthcare (Wu et al. 2024b). Some focus on particular domains; for example, in NLP, some focus on how causal inference can improve interpretability (Feder et al. 2022), or ways to explain (Danilevsky et al. 2020; Lyu, Apidianaki, and Callison-Burch 2024) or interpret (Madsen, Reddy, and Chandar 2022) neural NLP systems. Our survey is more specifically focused on studies that aim to understand NLP systems via their internal components—and even more specifically, those that do so via causal techniques such as interventions to those components.

Tools. Several libraries have recently been released to facilitate causal interpretability methods that involve interventions to model components. These tools can implicitly prioritize certain types of mediators over others. For instance, *pyvene* (Wu et al. 2024d) is designed specifically to aid in locating non-basis-aligned multidimensional subspaces via alignment search methods such as distributed alignment search and its successors (Geiger et al. 2024; Wu et al. 2023; Huang et al. 2024; Wu et al. 2024c). While it can also be used for other kinds of model interventions, this library could be particularly useful for those wishing to verify existing causal hypotheses. *TransformerLens* (Nanda and Bloom 2022) and libraries based on it (*Prisma*; Joseph 2023) are interpretability tools for examining Transformer-based neural networks. In these libraries, the interface is standardized across model architectures. This tends to encourage a focus on basis-aligned components such as neurons and attention heads, subspaces, and layers, as interventions to these mediators are natively supported. *NeuroX* (Dalvi, Sajjad, and Durrani 2023) similarly incentivizes neuron-level interpretability in particular. *NNsight* (Fiotto-Kaufman et al. 2025) and *Baukit* (Bau 2022) are more transparent interfaces that provide access to the underlying PyTorch model architecture, which allows for flexible modifications of the model’s computation graph. Due to different naming conventions across model developers, this more transparent access may make it harder to generalize basis-aligned intervention code across architectures at first, but research on both basis-aligned and non-basis-aligned mediators is more accessible under this paradigm.

Note that this survey is intended more as a scientific review of the field rather than a practical guide to using these tools, so we have mainly discussed these toolkits with respect to the mediators that they enable working with. Some surveys such as Ferrando et al. (2024); Rai et al. (2024) or code tutorials (Nanda and Bloom 2022; Wu et al. 2024d; Fiotto-Kaufman et al. 2025; Mohebbi et al. 2024) make hands-on practical introduction to particular methods their explicit purpose, without necessarily assuming (nor discussing the benefits nor drawbacks of) a particular categorization of methods, nor their units of analysis. See these surveys for more hands-on guides to implementing interpretability methods.